

Roll No.

TPE-712

**B. TECH. (PETROLEUM ENGINEERING)
(SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
RESERVOIR MANAGEMENT**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) What are the various techniques for identification of the prospect form oil and gas reservoir point of view ? Explain in detail at least *four* techniques. (CO1)

OR

- (b) How the various type of Geoscientific data is integrated for exploration and development of oil and gas reservoir ? Explain in details. (CO1)

2. (a) What are the essential reservoir parameters for describing the reservoir ? What are the various logging tools for their measurements ? (CO2)

OR

- (b) What are the various types of direct and indirect methods of reservoir analysis ? How do these methods help in study of reservoir engineering ? (CO2)

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(2)

3. (a) Define cubical packing, orthorhombic packing and rhombohedra packing of spheres from porosity point of view. How much will be the porosity contribution of these packings ? (CO2)

OR

- (b) What are the various logging tools for porosity estimation in the borehole ? What is the tool physics of all tools behind the porosity estimation ? (CO2)
4. (a) How many type models required for exploration and development of oil and gas field ? What are the various types of software uses for building these models ? (CO1, CO2)

OR

- (b) What is a sequential workflow for building a reservoir model ? Explain its various components in detail. (CO1, CO2)
5. (a) What is the difference between facies log and other logs ? What are the various methods for facies log population in 3D reservoir modeling ? (CO2)

OR

- (b) Compare deterministic and stochastic methods for Petrophysical modeling. Write Petrophysical modeling workflow with detail of each step. (CO2)